

IMEN266 Operations Research II

Chapter 7. Renewal Theory

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

- 1 Renewal Process
- 2 Limiting behavior
- 3 Wald's equation
- 4 Renewal reward process
- 5 Regenerative process

How to study this chapter (2026 format)

- **Slides** (this file): concepts, class-problem statements, and the worked derivations developed in class. No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP9 renewal process, renewal function, limit theorems, Wald · CP10 renewal reward, age/residual life, inspection paradox, regenerative and alternating renewal processes): in class the INCLASS version (fill the TODOs, `check()` answers instantly); at home the full version with solutions and experiments.
- **Companion** (gap-fill → full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 4**: A exam-style (no AI first) · B critique an AI solution · C stress-test the rate of convergence and the insensitivity claims by simulation · D prompt log.
- Class rhythm (75 min): quiz 5' → concepts → live notebook → wrap-up. Ch. 7 = weeks 9–11.
- Ground truth = analytic $\leftrightarrow$ simulation agreement, not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

Renewal Process

- Recall the Poisson process
- What if inter-event distribution is not exponential?
- The renewal process is a generalization of the Poisson process by relaxing the exponential assumption of inter-event times.

Definition

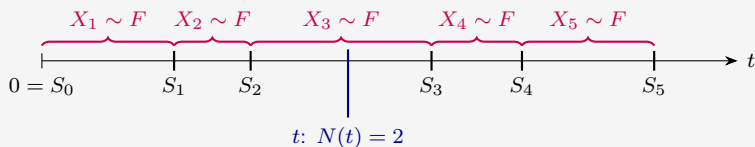
Suppose we have a sequence of iid non-negative random variables, $X_1, X_2, \dots$. Then, the counting process $\{N(t), t \geq 0\}$ is called a renewal process.

▷ Companion §1.1: S_n , $N(t)$, and the key identity $N(t) \geq n \Leftrightarrow S_n \leq t$ (Proof 1).

▷ Recall: the Poisson process is the counting process whose inter-event times are iid *exponential*; a renewal process keeps “iid” and drops “exponential”.

Renewal Process

The objects: gaps X_n , renewal epochs S_n , the count $N(t)$



$X_1, X_2, \dots$ iid with distribution F , $\mu = E[X_1]$; $S_n = X_1 + \dots + X_n$ is the time of the n -th renewal ($S_0 = 0$); $N(t) = \max\{n : S_n \leq t\}$ counts the renewals in $(0, t]$. The key identity, read off the picture:

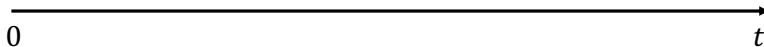
$$N(t) \geq n \iff S_n \leq t.$$

Two standing assumptions

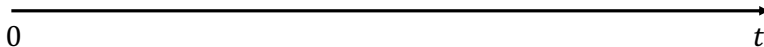
$F(0) = P(X_n = 0) < 1$, so $\mu > 0$ and, by the strong law, $S_n/n \rightarrow \mu > 0$: the epochs $S_n \rightarrow \infty$ and $N(t) < \infty$ for every t (only finitely many renewals in a finite interval). The Poisson process is the special case $F = \exp(\lambda)$.

Why renewal?

- Poisson process

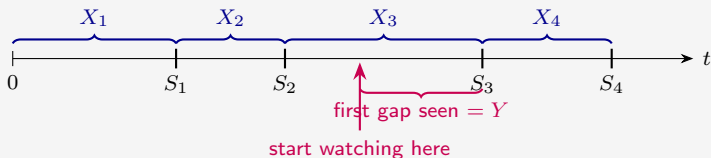


- Renewal process



Why renewal?

Where does the process restart?



Poisson process ($X_n \sim \exp(\lambda)$): start watching at *any* time — by memorylessness the first gap seen is again $\exp(\lambda)$, independent of the past: the Poisson process is renewed at every time point you choose.

Renewal process ($X_n \sim F$): from a non-event time the first gap seen is the *remaining* part of an interval; it does not have distribution F (it tends to be longer — inspection paradox). Only at an event time does the future look like a fresh copy: a renewal process is renewed at its event times $S_1, S_2, \dots$ only.

▷ Hence the age $A(t)$ and the remaining time $Y(t)$ at a non-renewal time t .

Distribution of $N(t)$

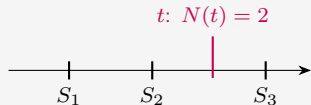
$$\begin{aligned}P(N(t) = n) &= P(N(t) \geq n) - P(N(t) \geq n+1) \\&= P(S_n \leq t) - P(S_{n+1} \leq t) \\&= P\left(\sum_{i=1}^n X_i \leq t\right) - P\left(\sum_{i=1}^{n+1} X_i \leq t\right) \\&= F_n(t) - F_{n+1}(t).\end{aligned}$$

- Example: Poisson process

▷ Companion Proof 1: $P(N(t) = n) = F_n(t) - F_{n+1}(t)$; for exponential gaps this recovers $e^{-\lambda t}(\lambda t)^n/n!$.

Distribution of $N(t)$

Why $P(N(t) \geq n) = P(S_n \leq t)$, and what F_n is



“at least n renewals by time t ” $\Leftrightarrow$ “the n -th renewal happened by t ”:

$$\{N(t) \geq n\} = \{S_n \leq t\}.$$

$F_n(t) = P(S_n \leq t)$ is the

n -fold convolution of F : $F_1 = F$,

$$F_{n+1}(t) = \int_0^t F_n(t-x) dF(x).$$

Example: exponential gaps give back the Poisson pmf

$S_n \sim \text{Erlang}(n, \lambda)$, so $F_n(t) = 1 - \sum_{i=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^i}{i!}$ (“fewer than n events by t ” is the complement), and $P(N(t) = n)$ is

$$F_n(t) - F_{n+1}(t) = \sum_{i=0}^n e^{-\lambda t} \frac{(\lambda t)^i}{i!} - \sum_{i=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^i}{i!} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

▷ General F : no closed form — CP9 computes F_n numerically and by simulation.

Renewal equation

Lemma

$$m(t) \equiv E[N(t)] = \sum_{n=1}^{\infty} F_n(t)$$

▷ Companion §1.2 and Proof 2: the renewal function $m(t) = \sum_n F_n(t)$; CP9 computes it three ways.

Renewal equation

Proof of $m(t) = \sum_{n \geq 1} F_n(t)$ — the tail-sum formula

For a nonnegative integer-valued random variable,
 $E[N] = \sum_{n=1}^{\infty} P(N \geq n)$ (the event $\{N \geq n\}$ is counted once for each $n \leq N$; continuous analogue: $E[X] = \int_0^{\infty} (1 - F(t)) dt$). Apply it to $N(t)$:

$$m(t) = E[N(t)] = \sum_{n=1}^{\infty} P(N(t) \geq n) = \sum_{n=1}^{\infty} P(S_n \leq t) = \sum_{n=1}^{\infty} F_n(t).$$

The renewal function

$m(t)$ is the **renewal function** of F : the mean number of renewals in $(0, t]$. It is finite for every t (Companion Proof 1); for the Poisson process $m(t) = \lambda t$, for a general F the sum of convolutions is rarely closed-form — hence the equation on the next slide.

▷ Tail-sum formulas return in Wald's equation and in the age process (§5).

Renewal equation

Lemma (Renewal equation)

$$m(t) = F(t) + \int_0^t m(t-x)dF(x)$$

▷ Companion Proof 2 (final-style): solving the renewal equation — uniform(0, 1) by differentiation, hyperexponential by Laplace transform.

Renewal equation

Proof: condition on the first renewal time X_1

$$m(t) = E[N(t)] = E[E[N(t) \mid X_1]] = \int_0^\infty E[N(t) \mid X_1 = x] dF(x).$$

If $X_1 = x \leq t$: one renewal at x , then a fresh renewal process on $(x, t]$, so $E[N(t) \mid X_1 = x] = 1 + m(t - x)$. If $x > t$: no renewal by t , so $E[N(t) \mid X_1 = x] = 0$. Hence

$$m(t) = \int_0^t (1 + m(t - x)) dF(x) + \int_t^\infty 0 dF(x) = F(t) + \int_0^t m(t - x) dF(x).$$

What the equation is good for

It determines m from F without any convolution: solve it on a time grid (CP9 §B), by Laplace transforms (Companion Proof 2), or via an ODE (next frame).

▷ After X_1 the process is a fresh copy — the standard first step.

Example: $F \sim \text{uniform}(0, 1)$

- Let the inter-event times be $\text{uniform}(0, 1)$, so that $F(t) = t$ and $dF(t) = dt$ for $0 \leq t \leq 1$.
- Find the renewal function $m(t)$ for $0 \leq t \leq 1$ from the renewal equation.

▷ Companion Proof 2(c); CP9 §B checks $m(t) = e^t - 1$ against the discretised renewal equation and simulation.

Example: $F \sim \text{uniform}(0, 1)$

Solve the renewal equation for $0 \leq t \leq 1$

For $0 \leq t \leq 1$: $F(t) = t$ and $dF(x) = dx$ on $[0, t]$, so

$$m(t) = t + \int_0^t m(t-x) dx = t + \int_0^t m(u) du \quad (u = t-x).$$

Differentiate: $m'(t) = 1 + m(t)$, $m(0) = 0$. Put $g = m + 1$: $g' = g$, $g(0) = 1$, so $g(t) = e^t$ and

$$m(t) = e^t - 1 \quad (0 \leq t \leq 1).$$

$$m(\tfrac{1}{2}) = e^{1/2} - 1 \approx 0.649, \quad m(1) = e - 1 \approx 1.718 \text{ (CP9 §B)}.$$

Beyond $t = 1$, and the long-run check

For $t > 1$ the integral only runs over $x \in [0, 1]$: $m(t) = 1 + \int_{t-1}^t m(u) du$ — piecewise, no longer one exponential (CP9 solves it on a grid).

Elementary renewal theorem: $m(t)/t \rightarrow 1/\mu = 2$; more precisely $m(t) = 2t - \frac{1}{3} + o(1)$ (Companion §1.2), and $m(1) = 1.718$ is already close to $2 - \frac{1}{3} = 1.667$.

Limit theorem

- Strong law of large numbers

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu = E[X_1] \text{ a.s.}$$

▷ Companion §1.3: almost-sure convergence (one sample path at a time) versus convergence in distribution (CLT).

Limit theorem

What “almost surely” says: one sample path at a time

$S_n/n = \frac{1}{n} \sum_{i=1}^n X_i$ is the running average. The strong law states

$$P\left(\left\{\omega : \lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} = \mu\right\}\right) = 1 :$$

for (almost) every realisation ω of the *whole sequence*, the running average converges to μ . Coin flips ($E[X] = \frac{1}{2}$): $\omega_1 = HHHH \dots$ has running average $\rightarrow 1$, $\omega_2 = TTTT \dots$ has $\rightarrow 0$ — such paths exist, but together they have probability 0; a typical path $HHTHT \dots$ converges to $\frac{1}{2}$.

Contrast: the central limit theorem is about distributions

$Z_n = \frac{S_n/n - \mu}{\sigma/\sqrt{n}} \Rightarrow N(0, 1)$ in distribution: the fluctuation of S_n/n

around μ has size $\sigma/\sqrt{n}$ and a Gaussian shape. Both are used here: the strong law for long-run *rates*, the CLT for the accuracy of simulation estimates.

Limit theorem

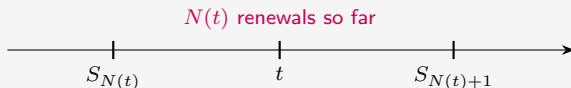
Lemma

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu} \text{ a.s.}$$

▷ Companion Proof 3: the sandwich $S_{N(t)}/N(t) \leq t/N(t) < S_{N(t)+1}/N(t)$; the long-run rate of a renewal process is $1/\mu$.

Limit theorem

Proof: sandwich t between two renewal epochs



By definition $S_{N(t)} \leq t < S_{N(t)+1}$. Divide by $N(t)$:

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)} = \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)}.$$

As $t \rightarrow \infty$, $N(t) \rightarrow \infty$ a.s. (every S_n is finite), so the strong law applies along $n = N(t)$: the left side $\rightarrow \mu$ and the right side $\rightarrow \mu \cdot 1$. Hence $t/N(t) \rightarrow \mu$ a.s., i.e.

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu} \quad \text{a.s.} \quad (\text{rate } 0 \text{ if } \mu = \infty).$$

▷ Only the mean μ enters the rate; the shape of F sets how fast $N(t)/t$ settles (CP9 §C, HW4 Part C).

Elementary renewal theorem

Theorem (Elementary renewal theorem)

$$\lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu}.$$

▷ Companion Proof 3: $m(t)/t \rightarrow 1/\mu$ needs Wald's equation (next section) — a.s. convergence alone does not give convergence of expectations.

Elementary renewal theorem

Why this is not a corollary of the previous lemma

$N(t)/t \rightarrow 1/\mu$ almost surely does *not* by itself give $E[N(t)]/t \rightarrow 1/\mu$ (a.s. convergence versus convergence of expectations). The proof (Companion Proof 3) uses Wald's equation of the next section:

$$E[S_{N(t)+1}] = \mu(m(t) + 1) \quad \text{and} \quad S_{N(t)+1} > t \quad \implies \quad \frac{m(t)}{t} > \frac{1}{\mu} - \frac{1}{t},$$

so $\liminf_t m(t)/t \geq 1/\mu$; the reverse inequality comes from truncating X at a level M .

Sanity check: the Poisson process

$m(t) = E[N(t)] = \lambda t$, so $\frac{m(t)}{t} = \lambda = \frac{1}{\mu}$ for every t — no limit needed.

For a general F the ratio only converges: for uniform(0,1) gaps, $m(1)/1 = e - 1 \approx 1.72$ versus $1/\mu = 2$.

▷ Both: the long-run renewal rate is $1/\mu$, per path and on average.

Class problem 9

1. Kim has a computer mouse that works with a single battery. Whenever the battery is out, Kim replaces it with a new one immediately. Given that the lifetime distribution of the battery is $\text{uniform}(50, 100)$, what is the average rate at which Kim replaces the battery in the long run?

► Colab notebook: CP9 §1 (battery replacement rate) [INCLASS] [full]

Class problem 9

1. Battery replacements: a renewal process with uniform(50, 100) lifetimes

$X_i \sim \text{uniform}(50, 100)$ (hours) is the lifetime of the i -th battery; replacement is immediate, so the replacement epochs are $S_n = X_1 + \cdots + X_n$ and $N(t)$ = number of replacements by time t : a renewal process with $\mu = E[X_1] = 75$ hours.

$$\text{replacement rate} = \lim_{t \rightarrow \infty} \frac{N(t)}{t} = \lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu} = \frac{1}{75} \text{ per hour}$$

≈ 0.0133 per hour: about 13.3 batteries per 1000 hours (CP9 §C).

What the answer does and does not depend on

Only the mean lifetime matters: uniform(50, 100), exponential with mean 75 and a deterministic 75-hour battery all give $1/75$. They differ in the *fluctuations*: deterministic lifetimes keep $N(t)/t$ within $1/t$ of $1/75$, the others converge like $\sigma/\sqrt{t}$ (CP9 §C).

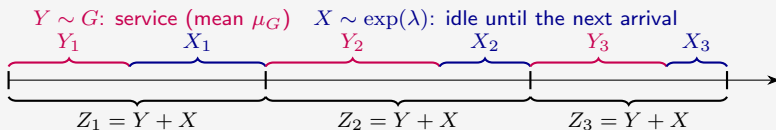
Class problem 9

2. Consider a bank with a single server. Customers arrive according to $PP(\lambda)$. Each customer enters the bank only when the server is free. Otherwise, the customer leaves without entering the bank. The service times of the server follow a general distribution with mean μ_G .
- (a) What is the customers' entering rate to the bank?
 - (b) What proportion of arriving customers actually enter the bank?

► Colab notebook: CP9 §2 (single-server bank) [INCLASS] [full]

Class problem 9

2. Bank with a single server: cycles = service + idle period



A cycle starts when a customer enters (server free): the service Y , then an idle period $X \sim \exp(\lambda)$ until the next arrival (memoryless, whatever happened before), so the cycles $Z_n = Y_n + X_n$ are iid: the entries form a renewal process with $E[Z] = \mu_G + 1/\lambda$.

$$(a) \text{ entering rate} = \lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{E[Z]} = \frac{\lambda}{1 + \lambda\mu_G} \text{ a.s.}$$

$$(b) \text{ proportion entering} = \frac{\text{entering rate}}{\text{arrival rate } \lambda} = \frac{1}{1 + \lambda\mu_G}$$

$\lambda = 4$ per hour, $\mu_G = 10$ min: 2.4 customers per hour enter, i.e. 60 % of the arrivals (CP9 §C).

Wald's equation

Definition (Stopping time)

The non-negative integer-valued random variable N is said to be a stopping time for a sequence of independent random variables $X_1, X_2, \dots$ if the event $\{N = n\}$ is independent of $X_{n+1}, X_{n+2}, \dots$

- Example: $X_1, X_2, \dots$ is a sequence of iid random variables with $P(X_i = 1) = P(X_i = -1) = 0.5$.

$$N \equiv \min\{n | X_1 + \dots + X_n = 1\}.$$

Is N a stopping time?

▷ Companion §1.4: a stopping time may look at the past, never at the future.

Wald's equation

From a fixed number of terms to a random one

Fixed n : $E[\sum_{i=1}^n X_i] = nE[X_1]$. Random N : $E[\sum_{i=1}^N X_i] = ?$ If N is *independent* of the sequence: $E[N]E[X_1]$ (condition on N). Wald's equation keeps this formula when N may depend on the X_i 's, provided "stop at n " looks only at $X_1, \dots, X_n$:

$\{N = n\}$ is determined by $X_1, \dots, X_n$ (a stopping time).

The example: $N = \min\{n : X_1 + \dots + X_n = 1\}$ with $P(X_i = \pm 1) = \frac{1}{2}$

$\{N = n\}$ = "the partial sums stayed below 1 up to step $n - 1$ and hit 1 at step n " — determined by $X_1, \dots, X_n$: **yes, a stopping time**. (Not one: "the last $n \leq 10$ with $X_n = 1$ ".)

▷ Caution: $E[N] = \infty$ here, so Wald does *not* apply:
 $E[S_N] = 1 \neq E[N]E[X_1]$.

Wald's equation

Theorem (Wald's equation)

If $X_1, X_2, \dots$ is a sequence of iid random variables with finite expectation and if N is a stopping time for this sequence such that $E[N] < \infty$, then

$$E \left[\sum_{n=1}^N X_n \right] = E[N] E[X].$$

▷ Companion Proof 4: Wald's equation via the indicators $I_n = 1\{N \geq n\}$, which depend only on $X_1, \dots, X_{n-1}$.

Wald's equation

Proof (1): write the random sum with indicators

$$\sum_{n=1}^N X_n = \sum_{n=1}^{\infty} X_n I_n, \quad I_n = \begin{cases} 1 & \text{if } n \leq N \\ 0 & \text{otherwise} \end{cases} = \mathbf{1}\{N \geq n\}.$$

Claim: I_n and X_n are independent.

$\{N \leq n-1\} = \{N=1\} \cup \{N=2\} \cup \dots \cup \{N=n-1\}$, and each $\{N=i\}$ is determined by $X_1, \dots, X_i$ (stopping time); therefore

$\{N \geq n\} = \{N \leq n-1\}^c$ is determined by $X_1, \dots, X_{n-1}$ — note the shift: “ $\geq n$ ” needs only the first $n-1$ terms — and by the independence of the sequence

$$I_n = \mathbf{1}\{N \geq n\} \text{ is independent of } X_n.$$

▷ This is the whole trick: a stopping time may use the past, and “still going at step n ” is a statement about the past of step n .

Wald's equation

Proof (2): take expectations term by term

$$\begin{aligned} E\left[\sum_{n=1}^N X_n\right] &= E\left[\sum_{n=1}^{\infty} X_n I_n\right] \stackrel{(i)}{=} \sum_{n=1}^{\infty} E[X_n I_n] \stackrel{(ii)}{=} \sum_{n=1}^{\infty} E[X_n] E[I_n] \\ &= E[X_1] \sum_{n=1}^{\infty} P(N \geq n) \stackrel{(iii)}{=} E[X_1] E[N]. \end{aligned}$$

(i) monotone convergence for $X_n \geq 0$ (dominated convergence in general); (ii) independence of X_n and I_n ; (iii) tail-sum formula.

Where $E[N] < \infty$ is used

The interchange in (i) for signed X_n requires

$\sum_n E|X_n| E[I_n] = E|X_1| E[N] < \infty$; without it the random sum need not have expectation $E[X]E[N]$ (random-walk example).

▷ Needed: iid terms with a finite mean, and a stopping rule with $E[N] < \infty$.

Wald's equation

Corollary

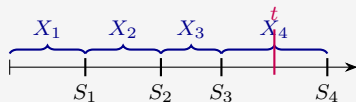
$$E[S_{N(t)+1}] = E[X] E[N(t) + 1].$$

▷ Companion Proof 4: $N(t)$ is not a stopping time, $N(t) + 1$ is;

$E[S_{N(t)+1}] = \mu(m(t) + 1)$ is the key to Proof 3.

Wald's equation

$N(t)$ is not a stopping time — $N(t) + 1$ is



$\{N(t) = 3\} = \{S_3 \leq t < S_4\}$ depends on X_4 , the term *after* the first three: $N(t)$ looks at the future — **not** a stopping time.

$\{N(t) + 1 = 4\} = \{S_3 \leq t < S_4\}$ is determined by the first four terms $X_1, \dots, X_4$:
 $N(t) + 1$ is a stopping time.

Hence the corollary

$E[N(t) + 1] = m(t) + 1 < \infty$, so Wald applies to the first $N(t) + 1$ gaps:

$$E[S_{N(t)+1}] = E\left[\sum_{n=1}^{N(t)+1} X_n\right] = E[X_1] E[N(t) + 1] = \mu(m(t) + 1).$$

▷ By contrast $E[S_{N(t)}] \neq \mu m(t)$ in general (CP9 §D checks both).

Renewal reward process

- Each time a renewal occurs, we receive a reward R .
- R_n is a reward earned at n^{th} renewal, independent of each other.

Theorem (Renewal reward process)

Define a stochastic process

$$R(t) = \sum_{n=1}^{N(t)} R_n.$$

If $E[R] < \infty$, $E[X] < \infty$, then

$$(a) \lim_{t \rightarrow \infty} \frac{R(t)}{t} = \frac{E[R]}{E[X]} \text{ a.s.}$$

$$(b) \lim_{t \rightarrow \infty} \frac{E[R(t)]}{t} = \frac{E[R]}{E[X]}.$$

Renewal reward process

Reading the theorem

$R(t) = \sum_{n=1}^{N(t)} R_n$ is the **cumulative reward until time t** . The pairs (X_n, R_n) are iid: a cycle's reward may depend on *its own* length, not on the other cycles. Proof of (a):

$$\frac{R(t)}{t} = \frac{\sum_{n=1}^{N(t)} R_n}{N(t)} \cdot \frac{N(t)}{t} \xrightarrow{t \rightarrow \infty} E[R] \cdot \frac{1}{E[X]} \quad \text{a.s.}$$

by the strong law ($N(t) \rightarrow \infty$) and the limit theorem; the partial last cycle adds at most $R_{N(t)+1}/t \rightarrow 0$.

$$\text{long-run reward rate} = \frac{E[\text{reward in one cycle}]}{E[\text{length of one cycle}]} \quad \text{never } E[R/X].$$

The recipe for every class problem (Companion §1.5, Proof 5)

(1) find a **renewal epoch** (the system restarts probabilistically); (2) $E[X]$ = expected cycle length; (3) $E[R]$ = expected reward or cost in one cycle; (4) answer = $E[R]/E[X]$.

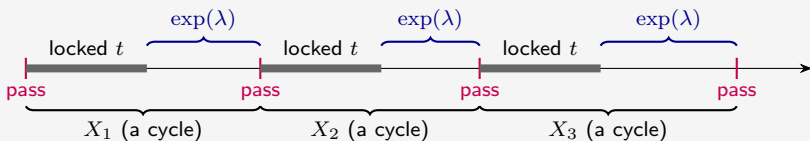
Class problem 9

3. Customers arrive to a single-server queueing system according to $PP(\lambda)$. The arriving customer must pass through a door to reach the server. Each time a customer passes through, the door is locked for the next t units of time. If the door is locked when a customer arrives, the customer is lost incurring a cost c . After passing through the door, if the server is free, the customer gets service. If not, the customer is lost incurring a cost K . If the service times follow $\exp(\mu)$, what is the average cost per unit time?

► Colab notebook: CP10 §3 (door with a lockout) [INCLASS] [full]

Class problem 9

3. The door: a cycle starts each time a customer passes through



After a customer passes, the door is locked for t ; every arrival during the lock is lost (cost c). After the lock opens, the next arrival comes after an $\exp(\lambda)$ time (Poisson arrivals, memoryless) and passes — a new cycle. The cycles are iid, so the passing epochs form a renewal process with

$$X = t + \exp(\lambda), \quad E[X] = t + \frac{1}{\lambda}.$$

▷ The cost K concerns the server: the passing customer is served if the server is free, lost if it is still busy. Next: the cost of one cycle.

Class problem 9

3. Expected cost per cycle $R = C_1 + C_2$, then divide by $E[X]$

C_1 = cost of the customers lost at the locked door: $\text{Poisson}(\lambda t)$ of them, so $E[C_1] = \lambda t c$.

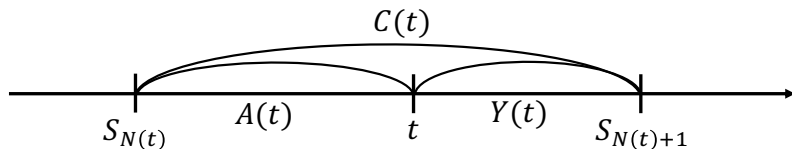
C_2 = cost K if the customer passing at the *end* of the cycle finds the server busy. $Y \sim \exp(\mu)$: remaining service time at the cycle start (memoryless — the same whether or not the first customer was served); $Z \sim \exp(\lambda)$: wait after the lock. Lost iff $Y > t + Z$:

$$\begin{aligned} E[C_2] &= K P(Y > t + Z) = K \int_0^\infty \lambda e^{-\lambda z} e^{-\mu(t+z)} dz \\ &= K e^{-\mu t} \int_0^\infty \lambda e^{-(\lambda+\mu)z} dz = K e^{-\mu t} \frac{\lambda}{\lambda + \mu}, \end{aligned}$$

$$\lim_{u \rightarrow \infty} \frac{E[R(u)]}{u} = \frac{E[R]}{E[X]} = \frac{\lambda t c + K e^{-\mu t} \lambda / (\lambda + \mu)}{t + 1/\lambda}.$$

$\lambda = 1/\text{min}$, $t = 1 \text{ min}$, $\mu = 0.5/\text{min}$, $c = 1$, $K = 5$:
 $(1 + 2.022)/2 \approx 1.51 \text{ per minute (CP10 §3)}.$

Age and remaining time



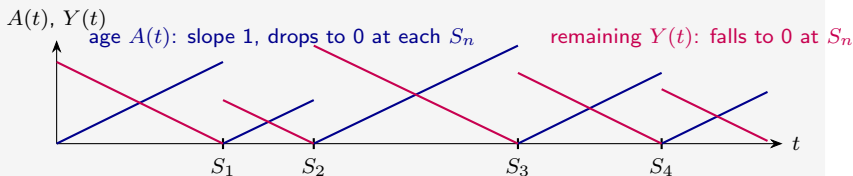
$$C(t) = A(t) + Y(t).$$

▷ Companion §1.6 and Proof 6: the time averages of $A(t)$ and $Y(t)$ are both $E[X^2]/(2E[X])$.

Age and remaining time

Three random variables attached to a time t

$A(t) = t - S_{N(t)}$ (**age**: time since the last renewal), $Y(t) = S_{N(t)+1} - t$ (**remaining** or residual life), $C(t) = X_{N(t)+1} = A(t) + Y(t)$ (the length of the interval that covers t).



Is $X_{N(t)+1} \sim F$? **No**: the interval that covers t is more likely to be a long one — it is sampled by *length* (inspection paradox, next frames).

▷ $A(t)$ itself never converges (a sawtooth); what converges is its *time average* — as the limiting probabilities of Ch. 5 are time fractions.

Age and remaining time

Time average of the age — a renewal reward argument

Reward of cycle n = area under the age curve during that cycle:

$R_n = \int_0^{X_n} s ds = X_n^2/2$ (iid, each depending on its own X_n). Split the integral at the last renewal before t :

$$\frac{1}{t} \int_0^t A(s) ds = \frac{\sum_{n=1}^{N(t)} R_n}{t} + \frac{\int_{S_{N(t)}}^t A(s) ds}{t} \xrightarrow{t \rightarrow \infty} \frac{E[X^2/2]}{E[X]} + 0,$$

the last term being at most $X_{N(t)+1}^2/(2t) \rightarrow 0$ (needs $E[X^2] < \infty$). The falling curve has the same area per cycle, and $C = A + Y$ adds up to $E[X^2]/E[X]$:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t A(s) ds = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t Y(s) ds = \frac{E[X^2]}{2E[X]} \quad \text{a.s.}$$

▷ $\exp(\lambda)$: $1/\lambda$; $\text{uniform}(0,1)$: $\frac{1}{3}$ (CP10 §B); $\text{deterministic } \mu$: $\mu/2$.

Inspection paradox

$$C(t) = A(t) + Y(t).$$

$$\lim_{t \rightarrow \infty} \frac{\int_0^t C(s) ds}{t} = \lim_{t \rightarrow \infty} \frac{\int_0^t A(s) ds}{t} + \lim_{t \rightarrow \infty} \frac{\int_0^t Y(s) ds}{t} = \frac{E[X^2]}{E[X]} \text{ a.s.}$$

$$\frac{E[X^2]}{E[X]} - E[X] = \frac{\mu^2 + \sigma^2}{\mu} - \mu = \frac{\sigma^2}{\mu} > 0.$$

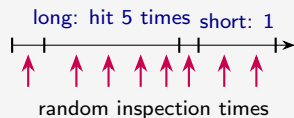
- Intuition: if we choose t equally likely, then it has higher probability to choose t in a longer interval.

▷ Companion Proof 6: the interval covering a random time is length-biased — density $xf(x)/E[X]$.

Inspection paradox

Why $E[\text{length of the interval covering } t] \geq E[X]$: sampling by length

Pick a time t “at random”. By renewal reward with $R = X \mathbf{1}\{X \in [x, x + dx]\}$, the fraction of the time axis covered by intervals of length in $[x, x + dx]$ is $x f(x) dx / E[X]$: the interval covering a random time has the length-biased density $x f(x) / E[X]$.



$$\text{Its mean: } \int_0^\infty x \cdot \frac{x f(x)}{E[X]} dx = \frac{E[X^2]}{E[X]} = E[X] + \frac{\sigma^2}{E[X]} \geq E[X]$$

with equality only for deterministic gaps ($\sigma = 0$).

The same bias, three ways

mean length seen = $E[X^2]/E[X]$ · mean age = mean remaining = $E[X^2]/(2E[X])$ · a Poisson arrival (“random time”) waits $E[X^2]/(2E[X])$ for the next renewal, not $E[X]/2$ — the bus stop of §4.

Inspection paradox

In daily life (Scientific American, J. Murtagh, 18 Jan 2024)

Trains. Headways alternate 4, 12, 4, 12, ... minutes (mean 8). A passenger arriving at a random time lands in a 12-minute gap with probability $12/16 = \frac{3}{4}$: the headway she experiences has mean $\frac{4 \cdot 4 + 12 \cdot 12}{16} = 10 \text{ min} = E[X^2]/E[X]$, and her mean wait is

$E[X^2]/(2E[X]) = 5 \text{ min}$, not $E[X]/2 = 4$.

Friends. Four people with 3, 2, 1, 2 friends (mean 2). Averaged over each person's friends, the friends have 1.67, 2.5, 3, 2.5 friends — mean $2.42 > 2$; counting every friendship from both ends, a friend has on average $\sum_i d_i^2 / \sum_i d_i = 18/8 = 2.25 = E[D^2]/E[D]$ friends: the same length-biased formula with the degree D in place of X . Popular people are counted by more friends, long intervals by more inspection times.

The call centre is always busy when you call — the same reason.

▷ Units selected “by being there” are biased toward large ones.

Regenerative process

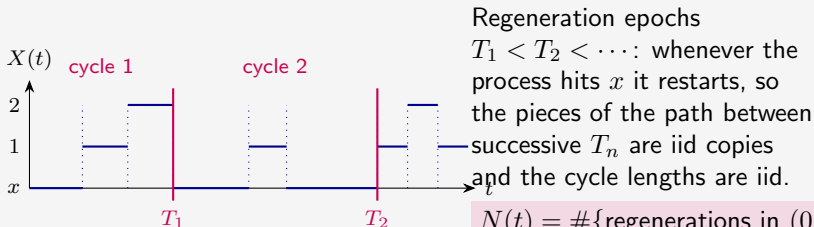
Definition

Let $\{X(t), t \geq 0\}$ be a stochastic process. Whenever $X(t)$ hits some values, let's say x , it is restarted. Then, $\{X(t), t \geq 0\}$ is called a regenerative process.

▷ Companion §1.7 and Proof 7: cycles are iid, so the regeneration epochs form a renewal process.

Regenerative process

Cycles: independent copies of the whole process



$N(t) = \#\{\text{regenerations in } (0, t]\}$
is a renewal process.

Examples

A renewal process itself (regenerates at each S_n); an ergodic CTMC (at each entry into a fixed state i); a queue (when a customer arrives to an empty system); the on/off system of the alternating renewal process. The Markov property is *not* required — only iid cycles.

Regenerative process

- Suppose $X(\tau_1) = x$, $X(\tau_2) = x$ for $\tau_1 < \tau_2$. Then, $X(\tau_1 + t)$ and $X(\tau_2 + t)$ are stochastically identical.
- If we consider a counting process that counts the number of regenerations until time t , it is a renewal process.
- For a regenerative process, we are interested in the fraction of time that the system stays in state j .
- Using the result of the renewal reward process,

$$\begin{aligned}\text{Fraction of time in state } j &= \frac{E[\text{time in state } j \text{ during a cycle}]}{E[T]} \\ &= \lim_{t \rightarrow \infty} P(X(t) = j).\end{aligned}$$

Regenerative process

Why: renewal reward with “time spent in j ” as the reward

Let R_n = time spent in state j during the n -th cycle (iid, each depending on its own cycle) and T = a cycle length. Then

$$\frac{1}{t} \int_0^t \mathbf{1}\{X(s) = j\} ds \xrightarrow[t \rightarrow \infty]{} \frac{E[\text{time in } j \text{ during a cycle}]}{E[T]} \quad \text{a.s.}$$

is the long-run fraction of time in j — exactly the renewal reward theorem.

Time fraction versus limiting probability

The slide's second equality, $\lim_{t \rightarrow \infty} P(X(t) = j) = E[\text{time in } j]/E[T]$, needs a *non-lattice* cycle length (not concentrated on multiples of some $d > 0$); for a lattice cycle $P(X(t) = j)$ may oscillate forever while the time fraction still exists. With a cycle density (this course) both agree.

▷ Ch. 5 took p_j from $p_Q = 0$ and deferred “time fraction” to here.

Example: ergodic CTMC

- We can think that whenever the system visits state i , the system restarts due to Markov property. So,

$$\text{Fraction of time in state } j = \frac{\text{mean time in state } j \text{ during a cycle}}{\text{mean time from } i \text{ to } i}.$$

▷ Companion Proof 7: $p_j = (1/\lambda_j)/m_j$ — the step that Ch.5 Proof 6(c) deferred to this chapter.

Example: ergodic CTMC

An ergodic CTMC regenerates at every entry into a fixed state i

Markov property + exponential sojourns: at each entry into i the past is forgotten, so the successive $i \rightarrow i$ excursions are iid cycles with mean length m_i (the mean return time). For any state j ,

$$p_j = \frac{E[\text{time in } j \text{ during an } i\text{-cycle}]}{m_i},$$

$$j = i: \quad p_i = \frac{E[\text{one sojourn in } i]}{m_i} \implies p_i = \frac{1/\lambda_i}{m_i}, \quad m_i = \frac{1}{\lambda_i p_i}.$$

This is the formula Ch. 5 Proof 6(c) and HW3 A3(v) used (“entries into i occur at rate $\lambda_i p_i$ ”).

Check: machine up/down (CP8 §1)

Cycle = one up time $\exp(\mu)$ + one repair $\exp(\lambda)$, starting at each repair completion: $p_{\text{up}} = \frac{1/\mu}{1/\mu + 1/\lambda} = \frac{\lambda}{\lambda + \mu}$, as from $pQ = 0$ — but the cycle formula only needs iid cycles, not exponential clocks (HW3 Part C).

Example: alternating renewal process

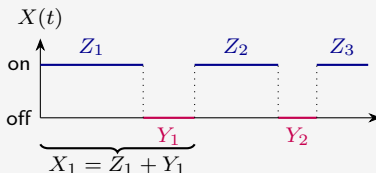
- Consider a system that changes states between on/off
- Z : random variable representing “on times”
- Y : random variable representing “off times”
- Z_n : n^{th} on time
- Y_n : n^{th} off time
- X_n : n^{th} cycle time, $Z_n + Y_n$

$$\text{Fraction of time that the system is on} = \frac{E[Z_n]}{E[X_n]}.$$

▷ Companion Proof 8 (final-style): on-fraction $E[Z]/(E[Z] + E[Y])$; the age process; insensitivity (HW3 Part C explained).

Example: alternating renewal process

On/off cycles



Cycle $X_n = Z_n + Y_n$ (on + off),
 $E[Z], E[Y] < \infty$; the pairs (Z_n, Y_n)
 are iid (Z_n, Y_n may depend on each
 other). Renewal reward with $R_n = Z_n$:

$$\text{fraction of time on} = \frac{E[Z]}{E[Z] + E[Y]}$$

($= \lim_t P(\text{on at } t)$ for a non-lattice
 cycle).

Examples

Machine with up times of mean m_u and repair times of mean m_r , any
 shapes: availability $m_u / (m_u + m_r)$. The bank of §2: busy fraction
 $\mu_G / (\mu_G + 1/\lambda)$. The premium rates of §5: "on" = paying r_1 .

Example: alternating renewal process

Final-style: the fraction of time the age is at most c

Call the system *on* at time t if $A(t) \leq c$. In a renewal interval of length X the age is $\leq c$ for $\min(X, c)$ time units: an alternating renewal process with on-time $Z = \min(X, c)$ and cycle X . The tail formula gives $E[\min(X, c)] = \int_0^\infty P(\min(X, c) > x) dx = \int_0^c (1 - F(x)) dx$, so

$$\text{fraction of time with } A(t) \leq c = \frac{\int_0^c (1 - F(x)) dx}{E[X]} =: F_e(c),$$

the **equilibrium distribution** of F (the same for $Y(t)$, by symmetry of the cycle).

Checks

$F = \exp(\lambda)$: $F_e(c) = \lambda \int_0^c e^{-\lambda x} dx = F(c)$ — the age of a Poisson process is again $\exp(\lambda)$. Uniform(0, 1): $F_e(c) = 2c - c^2$; the age is $\leq \frac{1}{4}$ for 43.75% of the time although only 25% of the intervals are that short (an age is measured *inside* an interval).

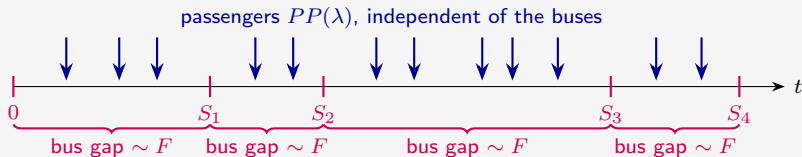
Class problem 9

4. Consider a bus stop. Passengers arrive to the bus stop according to $PP(\lambda)$. Buses arrive according to a renewal process with inter-arrival distribution F . Passengers and buses arrive to the bus stop independently.
- (a) the average number of people at the bus stop in the long run.
 - (b) the average amount of time that a passenger waits in the long run.

► Colab notebook: CP10 §4 (bus stop) [INCLASS] [full]

Class problem 9

4(a). Number waiting: reward = passenger-minutes accumulated in a bus cycle



Buses are the renewal epochs ($X \sim F$); each bus takes everyone waiting. In a cycle the number waiting s time units after the bus is a Poisson count $N_p(s)$ with $E[N_p(s)] = \lambda s$; the reward is $R = \int_0^X N_p(s) ds$:

$$\begin{aligned} E[R] &= \int_0^\infty E\left[\int_0^x N_p(s) ds\right] dF(x) \stackrel{\text{Fubini}}{=} \int_0^\infty \int_0^x \lambda s ds dF(x) \\ &= \int_0^\infty \frac{\lambda x^2}{2} dF(x) = \frac{\lambda E[X^2]}{2} \end{aligned}$$

$$L = \text{long-run average number waiting} = \frac{E[R]}{E[X]} = \frac{\lambda E[X^2]}{2E[X]}$$

Class problem 9

4(b). A passenger's wait: the remaining time seen by a Poisson arrival

Given one passenger arrival in $(0, t]$, its time is uniform on $(0, t]$ (Poisson property), independent of the buses: a passenger arrives at a “random time” of the bus process and waits the *remaining* time until the next bus, so her long-run average wait is the time average of Y :

$$W = \frac{E[X^2]}{2E[X]} \quad (\geq E[X]/2), \quad L = \frac{\lambda E[X^2]}{2E[X]} = \lambda W \quad (\text{Little's law, Ch. 8}).$$

Numbers: $\lambda = 2$ passengers/min, three bus schedules with mean headway 10 min

bus gaps F	$E[X^2]$	$W = E[X^2]/(2E[X])$	$L = \lambda W$
deterministic 10	100	5.0 min	10
uniform(5, 15)	108.3	5.42 min	10.8
exponential, mean 10	200	10.0 min	20

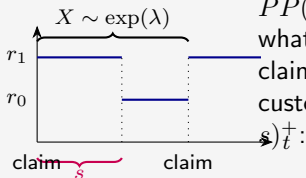
Class problem 9

5. The rate a certain insurance company charges its policyholders (customers) alternates between r_1 and r_0 . A new customer is initially charged at a rate r_1 per year. When a customer paying at rate r_1 has made no claims for s consecutive years, the rate becomes r_0 per year. The rate remains at r_0 until a claim is made, at which time it reverts to r_1 . If a customer makes claims according to $PP(\lambda)$,
- (a) calculate the proportion of time that the customer pays at rates r_1 and r_0 in the long run.
 - (b) obtain the long-run average amount paid per year.

► Colab notebook: CP10 §5 (insurance premiums) [INCLASS] [full]

Class problem 9

5. Premium rate: cycles from claim to claim



A claim resets the rate to r_1 ; claims form a $PP(\lambda)$, so the time to the next claim is $\exp(\lambda)$ whatever happened before: **cycle** = claim-to-claim interval $X \sim \exp(\lambda)$, $E[X] = 1/\lambda$. The customer pays r_1 for $\min(s, X)$ and r_0 for $(X - s)^+$:

$$E[\min(s, X)] = \int_0^s x \lambda e^{-\lambda x} dx + s e^{-\lambda s} = \frac{1}{\lambda} (1 - e^{-\lambda s}).$$

$$(a) \text{ at } r_1 : P_1 = \frac{E[\min(s, X)]}{E[X]} = \frac{(1 - e^{-\lambda s})/\lambda}{1/\lambda} = 1 - e^{-\lambda s},$$

$$\text{at } r_0 : P_0 = 1 - P_1 = e^{-\lambda s};$$

$$(b) \text{ average per year} = r_0 P_0 + r_1 P_1 = r_0 e^{-\lambda s} + r_1 (1 - e^{-\lambda s}).$$

$\lambda = 0.2/\text{yr}$, $s = 3 \text{ yr}$, $r_1 = 1000$, $r_0 = 600$: $P_0 = e^{-0.6} \approx 0.549$, average ≈ 780.5 per year (CP10 §5).